

Simple Python Program – Neural Network for Circular Data

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_circles
from sklearn.neural_network import MLPClassifier

# Create circular data
X, y = make_circles(n_samples=300, noise=0.1, factor=0.5)

# Create neural network
model = MLPClassifier(
    hidden_layer_sizes=(2, 2), # 2 hidden layers, 2 neurons each
    activation='tanh',
    solver='sgd',
    learning_rate_init=0.1,
    max_iter=1000,
    random_state=1
)

# Train the network
model.fit(X, y)

# Predict
y_pred = model.predict(X)

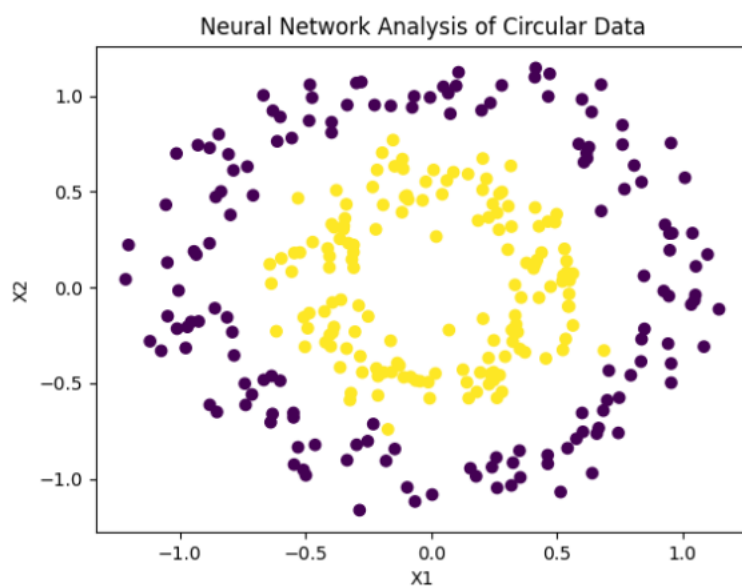
# Accuracy
accuracy = model.score(X, y)

print("Learning Rate:", 0.1)
print("Activation:", "Tanh")
print("Hidden Layers:", 2)
print("Hidden Neurons:", 2)
print("Accuracy:", accuracy)
```

```
# Plot circular data
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Neural Network Analysis of Circular Data")
plt.xlabel("X1")
plt.ylabel("X2")
plt.show()
```

OUTPUT:

```
Run started
Initializing environment
Installing packages
Running code
Learning Rate: 0.1
Activation: Tanh
Hidden Layers: 2
Hidden Neurons: 2
Accuracy: 0.8766666666666667
```



Run completed in 29519.70000001788ms